Swati Aggrawal

swatiagg357@gmail.com | +91-7011930926 | github.com/swatiaggrawal | linkedin.com/in/swatiaggrawal

RESEARCH PROFILE

M.Sc. Computer Science graduate (AI/ML specialization) from Sapienza Università di Roma with a peer-reviewed IEEE publication and a master's thesis on continual learning for Vision Transformers, achieving a ~97% relative reduction in catastrophic forgetting versus a CNN baseline. Combines research rigor in machine learning and algorithmic evaluation with hands-on engineering — building, benchmarking, and deploying ML systems in Python and C++ on AWS. Seeking a Research Sciences internship to contribute to applied research in machine intelligence.

SKILLS

- **Research & ML:** Machine Learning, Deep Learning, Computer Vision, Vision Transformers, Continual Learning, Transfer Learning, Algorithm Design & Analysis, Experimental Evaluation, PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers
- **Programming:** Python, C++, Java, SQL, Bash
- **Systems & Cloud:** AWS (ECS, Fargate, CloudWatch), Docker, Linux, FastAPI, REST APIs, Git
- **Generative AI:** LLMs, Retrieval-Augmented Generation (RAG), AI Agents, Vector Databases (Pinecone), Gemini API, Prompt Engineering

RESEARCH EXPERIENCE

Researcher — Vision Lab, Sapienza Università di Roma (Master's Thesis)

Nov 2025 – May 2026

Advisor: Prof. Marco Raoul Marini | https://github.com/swatiaggrawal/Thesis-Vision-Transformers

- Researched continual learning for computer vision, designing and running controlled experiments to isolate the effect of adapter-based fine-tuning and Elastic Weight Consolidation (EWC) on catastrophic forgetting.
- Built PyTorch training and evaluation pipelines benchmarking accuracy, forgetting, and computational efficiency across sequential classification tasks on Split CIFAR-100 (5 sequential tasks).
- **Achieved 90.59% average accuracy with only 2.01% average forgetting**, a ~97% relative reduction in catastrophic forgetting versus a naive ViT fine-tuning baseline (41.91% accuracy) and a CNN/ResNet-18 baseline (24.22% accuracy).
- Benchmarked against ResNet-18 and naive ViT fine-tuning baselines, validating results via cross-seed evaluation.
- Collaborated with researchers to document findings and compare state-of-the-art continual learning approaches.

Technologies: Python, PyTorch, Vision Transformers, Elastic Weight Consolidation, Computer Vision

PUBLICATION

Machine Learning for Air Pollution — IEEE ViTECoN

2023

doi.org/10.1109/ViTECoN58111.2023.10157028

- Authored a peer-reviewed IEEE survey paper reviewing six ML models (ANN, GA-ANN, Random Forest, Decision Tree, SVM, LS-SVM) used for air quality prediction, synthesizing findings from 35+ cited studies
- Reviewed air quality evaluation standards, including the six NAAQS criteria pollutants and their primary/secondary thresholds
- Analyzed prediction-based (classification/regression) and knowledge-discovery (clustering) learning paradigms applied in air pollution epidemiology
- Synthesized recent advances across ~12 published studies (2018–2022), covering methods like LightGBM, LSTM with genetic algorithms, and hybrid interpretable models
- Identified future research directions, including real-time IoT-integrated air quality monitoring and the shift toward deep learning approaches

APPLIED RESEARCH & ENGINEERING PROJECTS

LossyTextCompressor — Semantic Embedding Quantization

github.com/swatiaggrawal/LossyTextCompressor

- Studied the trade-off between compression rate and semantic preservation by generating and quantizing transformer text embeddings, then reconstructing text from compressed vectors.
- Evaluated reconstruction quality across compression levels using ROUGE-L and BLEU, quantifying the accuracy/efficiency frontier for embedding-based text compression.

Technologies: PyTorch, GTR-T5, Vec2Text, Quantization, ROUGE-L, BLEU

Knowledge Copilot — Grounded RAG Chatbot

github.com/swatiaggrawal/RAG-Chatbot | Live demo: rag-chatbot-new.onrender.com

- Built an end-to-end retrieval-augmented generation (RAG) chatbot that answers questions over FastAPI's official documentation, with every response grounded in retrieved source text and returned with per-answer citations to prevent hallucination.
- Architected a modular four-stage pipeline (ingestion, embedding, retrieval, generation), each independently runnable and testable via CLI, isolating failures to a specific stage.
- Implemented recursive paragraph/sentence-aware chunking with Unicode normalization to fix PDF ligature artifacts, improving embedding quality over naive fixed-size character splitting.
- Integrated Gemini API embeddings (task-type-aware asymmetric embeddings for queries vs. documents) with Pinecone as a managed vector store, removing a local sentence-transformers/torch dependency to fit the deployed backend within a 512MB free-tier RAM limit.
- Built a FastAPI backend exposing a /chat endpoint and a build-free React (CDN) frontend displaying source citations per answer; implemented self-indexing startup logic that auto-runs the ingest→embed→store pipeline on a fresh deploy.

Technologies: Python, FastAPI, React, Gemini API, Pinecone, RAG, LLMs, Vector Databases

DocDelta — AI-Powered Document Intelligence & Comparison Platform

github.com/swatiaggrawal/DocDelta

- Developed an end-to-end semantic document comparison API using FastAPI and Sentence Transformers (all-MiniLM-L6-v2), detecting and classifying sentence-level changes (added/removed/modified, numerical vs. wording) beyond traditional character-level diffs.
- Designed scalable embedding and cosine-similarity retrieval pipelines, achieving ~840ms warm-request latency on CPU-only inference.
- Containerized the application with Docker and deployed on AWS ECS/Fargate with CloudWatch monitoring for latency and error tracking.
- Built REST APIs with async request handling and validation, optimizing container size (CPU-only PyTorch builds) and eliminating cold-start overhead via model preloading.

Technologies: Python, FastAPI, Sentence Transformers, PyTorch, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision

github.com/swatiaggrawal/Metal-Defect-Detection

- Built a custom CNN (TensorFlow/Keras) plus SVM and Random Forest classifiers (HOG features) for automated industrial metal surface defect detection across 6 defect classes on the NEU dataset.
- Built image preprocessing, augmentation, training, and evaluation pipelines, tuning classical models via GridSearchCV.
- **Achieved 97.22% test accuracy** with the CNN versus 86.11% for both SVM and Random Forest — an ~11-point improvement demonstrating the value of learned spatial features over hand-crafted descriptors.
- Evaluated models using accuracy, confusion matrices, precision, recall, and F1-score.

Technologies: Python, TensorFlow, Keras, OpenCV, Scikit-Learn

EDUCATION

Sapienza Università di Roma — M.Sc. Computer Science 2026

- Relevant Coursework: Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- Thesis: Evaluating Pretrained Vision Transformers Under Continual Learning

Sharda University — B.Tech, CSE with AI and ML 2023

- Relevant Coursework: Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- Awarded merit-based scholarship (Top 5–10% of cohort)

Serial No.: 124818

Enrollment No.: 2019003547

SHARDA UNIVERSITY

Beyond Boundaries

On the recommendations of the Academic Council of

SHARDA UNIVERSITY

Greater Noida, Uttar Pradesh, India

Swati Aggrawal

having successfully completed the prescribed requirement

in **May, 2023,**

has been conferred the degree of

Bachelor of Technology

(Computer Science & Engineering with specialization in Artificial Intelligence & Machine Learning)

in **First Division with Honours.**

Registrar

Vice-Chancellor

Prepared By: Akumar

Checked By: PK

Verified By:

Date: 26/07/2023



This Document can be verified at
https://verification.sharda.ac.in/

Transcript No: SET-2139

SHARDA UNIVERSITY

Academic Transcript



Roll No. : 190101312
Name : SWATI AGGRAWAL

Mother's Name: ANSHU
Father's Name : RAKESH KUMAR

System ID : 2019003547
School: SHARDA SCHOOL OF ENGINEERING AND TECHNOLOGY
Programme: B.TECH.
Branch: COMPUTER SCIENCE & ENGINEERING WITH SPECIALIZATION IN ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Course Code	Course Name	Grade	Credits Earned
Semester - 1			
CHY161	ENGINEERING CHEMISTRY LAB	A+	1
PHY162	PHYSICS LAB-II	A+	1
CHY111	ENGINEERING CHEMISTRY	O	3
CSE113	PROGRAMMING FOR PROBLEM SOLVING	O	3
CSP101	INTRODUCTION TO COMPUTER SCIENCE AND ENGINEERING	A	1
HMM111	VALUES AND ETHICS	O	2
MTH142	CALCULUS AND ABSTRACT ALGEBRA	O	4
PHY117	SEMICONDUCTOR PHYSICS	O	3
ARP101	COMMUNICATIVE ENGLISH-I	A+	2
CSP113	PROGRAMMING FOR PROBLEM SOLVING LAB	O	1
MEP106	COMPUTER AIDED DESIGN AND DRAFTING LAB	O	1.5
Credits Earned : 22.5	***SGPA : 9.73***	***Total Credits Earned : 22.5***	
Semester - 2			
PHY161	PHYSICS LAB-I	A	1
CSA102	INTRODUCTION TO AI AND ML	O	2
CSE114	APPLICATION BASED PROGRAMMING IN PYTHON	O	3
EEE112	PRINCIPLES OF ELECTRICAL AND ELECTRONICS ENGINEERING	O	3
EVS112	ENVIRONMENTAL STUDIES	O	3
MTH145	PROBABILITY AND STATISTICS	O	4
PHY116	ENGINEERING PHYSICS	O	3
ARP102	COMMUNICATIVE ENGLISH-II	A	2
CSP103	MULTIMEDIA APPLICATION LAB	O	1
CSP114	APPLICATION BASED PROGRAMMING IN PYTHON LAB	A+	1
EEP112	PRINCIPLES OF ELECTRICAL AND ELECTRONICS ENGINEERING LAB	O	1
MEP105	MECHANICAL WORKSHOP	A	1.5
Credits Earned : 25.5	***SGPA : 9.61***	***Total Credits Earned : 48***	***CGPA : 9.67***
Semester - 3			
CSA202	CONCEPT OF MACHINE LEARNING	A+	3
CSE242	DATA STRUCTURES	B+	3
CSE243	OBJECT ORIENTED PROGRAMMING USING JAVA	A+	3
CSE244	PRINCIPLES OF OPERATING SYSTEMS	A	3
CSE245	DISCRETE STRUCTURES	A	4
CSE247	COMPUTER ORGANIZATION AND ARCHITECTURE	A+	3
ARP203	APTITUDE REASONING AND BUSINESS COMMUNICATION SKILLS-BASIC	A+	2
CAL201	CONCEPT OF MACHINE LEARNING LAB	A	1
CSP242	DATA STRUCTURES LAB	A+	1
CSP243	OBJECT ORIENTED PROGRAMMING USING JAVA LAB	A	1
CSP244	PRINCIPLES OF OPERATING SYSTEMS LAB	A+	1
CSP251	PROJECT BASED LEARNING (PBL)-I	A+	1
CSP294	SUMMER INTERNSHIP-I	A+	1
Credits Earned : 27	***SGPA : 8.44***	***Total Credits Earned : 75***	***CGPA : 9.23***
Semester - 4			
BTY223	INTRODUCTION TO BIOLOGY FOR ENGINEERS	A+	2
CSA203	CONCEPTS OF NEURAL NETWORKS	O	3
CSE011	MATHEMATICAL TECHNIQUES	A	3
CSE249	DATA BASE MANAGEMENT SYSTEM	A	3
CSE250	THEORY OF COMPUTATION AND COMPILER DESIGN	A+	4
CSE252	COMPUTER NETWORKS	O	3
MOO202	MANAGING CHANGE IN ORGANIZATIONS	A+	2
ARP204	APTITUDE REASONING AND BUSINESS COMMUNICATION SKILLS-INTERMEDIATE	A+	2
CSP249	DATA BASE MANAGEMENT SYSTEM LAB	A+	1
CSP250	THEORY OF COMPUTATION AND COMPILER DESIGN LAB	A+	1
CSP252	COMPUTER NETWORKS LAB	O	1
CSP298	PROJECT BASED LEARNING (PBL)-II	O	1
Credits Earned : 26	***SGPA : 9.08***	***Total Credits Earned : 101***	***CGPA : 9.19***

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Transcript No: **SET-2139**

Course Code	Course Name		Grade	Credits Earned
Semester - 5				
CSA022	INTRODUCTION TO CLOUD COMPUTING WITH MACHINE LEARNING		O	3
CSA301	SOFT COMPUTING		O	3
CSE350	DESIGN AND ANALYSIS OF ALGORITHM		O	4
CSE351	SOFTWARE ENGINEERING AND TESTING METHODOLOGIES		O	3
MOO310	INTRODUCTION TO INDUSTRY 4.0 AND INDUSTRIAL INTERNET OF THINGS		B+	3
ARP305	PERSONALITY DEVELOPMENT AND DECISION MAKING SKILLS		B	2
CSP350	DESIGN AND ANALYSIS OF ALGORITHM LAB		O	1
CSP351	PROJECT BASED LEARNING (PBL) -3		A+	1
CSP394	SUMMER INTERNSHIP-II		O	1
CSP395	TECHNICAL SKILL ENHANCEMENT COURSE-1 SIMULATION LAB		A	1
ECC301	COMMUNITY CONNECT		P	2
Credits Earned : 24	***SGPA : 8.67***	***Total Credits Earned : 125***	***CGPA : 9.09***	
Semester - 6				
CSA302	PATTERN RECOGNITION		A+	3
CSA303	DEEP LEARNING AND ITS APPLICATIONS		B+	3
CSE031	DIGITAL IMAGE PROCESSING		O	3
CSE041	SOFTWARE PROJECT MANAGEMENT		O	3
CSE354	WEB TECHNOLOGIES		O	2
HMM305	MANAGEMENT FOR ENGINEERS		A+	3
MEC344	NON CONVENTIONAL ENERGY		A	3
ARP306	CAMPUS TO CORPORATE		A	2
CAL302	PATTERN RECOGNITION LAB		A	1
CAL303	DEEP LEARNING AND ITS APPLICATIONS LAB		A+	1
CSP352	WEB TECHNOLOGIES LAB		A+	1
CSP392	PROJECT BASED LEARNING (PBL)-4		A	1
CSP396	TECHNICAL SKILL ENHANCEMENT COURSE-2 (APPLICATION DEVELOPMENT LAB)		O	1
Credits Earned : 27	***SGPA : 8.85***	***Total Credits Earned : 152***	***CGPA : 9.05***	
Semester - 7				
CSA401	COMPUTER VISION		O	3
CSE051	WIRELESS NETWORKS		O	3
CSE062	MOBILE COMPUTING		A	3
ONT700	SOFTWARE TESTING		B+	3
CAL401	COMPUTER VISION LAB		A+	1
CAL402	APPLICATIONS OF AIML IN HEALTHCARE/ ICT/ COMPUTER NETWORKS LAB		A+	3
CSP499	SUMMER INTERNSHIP-III		A+	1
CSP495	MAJOR PROJECT- 1		A+	3
Credits Earned : 20	***SGPA : 8.85***	***Total Credits Earned : 172***	***CGPA : 9.02***	
Semester - 8				
CSP496	MAJOR PROJECT		B+	8
Credits Earned : 8	***SGPA : 7.00***	***Total Credits Earned : 180***	***CGPA : 8.93***	

The Student has completed the programme.

Date of Issue: **27-06-2023**

Seal :

SHARDA UNIVERSITY COE

Prepared By:

Verified By:

For **Controller of Examinations**

SAPIENZA
UNIVERSITÀ DI ROMA

DEGREE CERTIFICATE

It is certified that **SWATI AGGRAWAL**, ID Number 2118031, born in GHAZIABAD, REPUBBLICA dell' INDIA on 13/03/2001, Tax ID code GGRSWT01C53Z222O gained at this University, on 26/05/2026 the master's degree in **COMPUTER SCIENCE - INFORMATICA [LM - Ordin. 2019 - Corso erogato in Lingua Inglese] (classe LM-18)**, Faculty of INGEGNERIA DELL'INFORMAZIONE, INFORMATICA E STATISTICA with grade **91/110**.

The statutory duration of the course is 2 academic years.

The present certificate, consists of 1 pages:

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- Is only valid with a revenue stamp for legally authorised purposes.

REVENUE STAMP
€ 16.00

Manager, Student Affairs Area
Dott.ssa Raffaela Iovane

Digital stamp Sapienza University of Rome.

To verify the digital stamp use the Decoder2DPlus software available at
`http://www.uniroma1.it/didattica/regolamenti/timbro-digitale`
Handwritten signature omitted in accordance with Article 3 of D. Lgs. No. 39/1993.

SAPIENZA
UNIVERSITÀ DI ROMA

TRANSCRIPT OF RECORDS

It is certified that **SWATI AGGRAWAL**, ID Number 2118031, born in GHAZIABAD, REPUBBLICA dell' INDIA on 13/03/2001, Tax ID code GGRSWT01C53Z222O is enrolled at this University for Academic Year 2023/2024 in the master's degree programme on **COMPUTER SCIENCE - INFORMATICA [LM - Ordin. 2019 - Corso erogato in Lingua Inglese] (classe LM-18)**, Faculty of INGEGNERIA DELL'INFORMAZIONE, INFORMATICA E STATISTICA.

During Academic Year 2025/2026 the student is enrolled in the 1° year fuori corso master's degree in **COMPUTER SCIENCE - INFORMATICA [LM - Ordin. 2019 - Corso erogato in Lingua Inglese] (classe LM-18)** , Faculty of INGEGNERIA DELL'INFORMAZIONE, INFORMATICA E STATISTICA.

Profit exams and educational activities

N°	Exam Code, Name and SSD	Date	Grade	Credits/Year
1	1047640 NETWORK ALGORITHMS (INF/01)	16/01/2024	18/30	c: 6
2	1041764 BIG DATA COMPUTING (INF/01)	07/02/2024	18/30	c: 6
3	1041792 BIOMETRIC SYSTEMS (INF/01)	20/02/2024	22/30	c: 6
4	1038141 NATURAL LANGUAGE PROCESSING (INF/01)	13/06/2024	27/30	c: 6
5	1047643 TOPICS IN PHYSICS (FIS/01)	19/06/2024	30/30	c: 6
6	1047630 HUMAN COMPUTER INTERACTION ON THE WEB (INF/01)	17/07/2024	18/30	c: 6
7	1047618 COMPUTER VISION (INF/01)	13/01/2025	28/30	c: 6
8	10596281 AUTONOMOUS NETWORKING (INF/01)	20/01/2025	26/30	c: 6
9	1047624 DISTRIBUTED SYSTEMS (INF/01)	28/01/2025	22/30	c: 6
10	1055050 RISK MANAGEMENT (SECS-P/11)	04/02/2025	26/30	c: 6
11	10593236 DEEP LEARNING AND APPLIED ARTIFICIAL INTELLIGENCE (INF/01)	13/02/2025	25/30	c: 6
12	1047619 CONCURRENT SYSTEMS (INF/01)	03/06/2025	20/30	c: 6
13	1047629 GRAPH THEORY (INF/01)	04/06/2025	21/30	c: 6
14	AAF1246 ADDITIONAL EDUCATIONAL ACTIVITIES (-)	09/06/2025	qualified	c: 6
15	AAF1034 FINAL EXAM (-)	-non disp.	qualified	c: 36

Passed exams: 15
Total credits: 120

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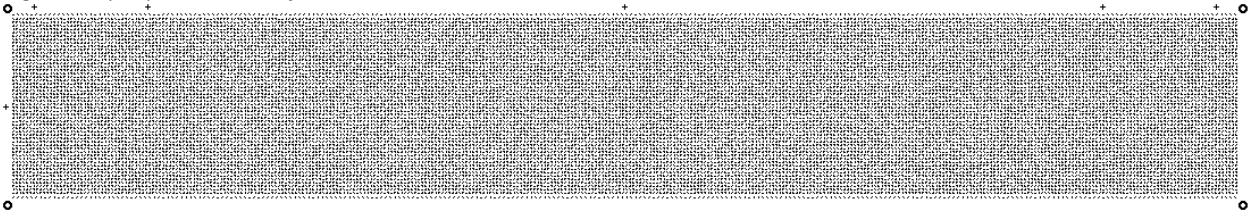
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Manager, Student Affairs Area
Dott.ssa Raffaela Iovane

Digital stamp Sapienza University of Rome.



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`http://www.uniroma1.it/didattica/regolamenti/timbro-digitale`
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